

Semantic Search Issue and Proposed Solution

The Problem

The current search system is effectively acting as a 'Keyword Search', even though it is built on a vector embeddings model.

This is happening because of recent updates made to prevent completely out-of-domain queries (e.g., searching for "Bill Gates" and getting Pakistan politics). To stop these false positives, **very strict keyword-presence checks and LLM-based validation rules** were added.

Specifically:

1. **Strict LLM Validation:** The AI is explicitly instructed to assign a score of 0-3 if the search topic or person is "NOT mentioned in the result". Any result scoring under 0.3 is automatically filtered out.
2. **Keyword Presence Penalty:** A penalty is artificially subtracted from the search score if the exact word isn't found in the text.

Because of these rigid rules, valid semantic matches are being thrown out. For example, if you search for "Constitution of Pakistan", the system might find a highly relevant video discussing "laws, amendments, and legal rights". However, because the exact word "Constitution" is missing from the transcript, the LLM gives it a low score and drops it entirely.

The Solution

To restore true meaning-based search while still protecting against completely irrelevant queries, we need to relax the strict keyword-matching prompts in the main testing file: **Clip matter py embedding/embeddings_test.py**.

1. Relax the LLM Reranker Prompt:

Instead of instructing the AI to penalize missing words, we will instruct it to focus on *meaning*:
"If the result discusses the exact meaning, concept, or underlying intent of the query, give it a high score (7-10), even if the specific words aren't used. Only give a low score (0-3) if the meaning is completely unrelated."

2. Adjust the Pre-Validation Prompt:

We will update the `validate_query_relevance` function prompt to explicitly tell the AI:
"Do not require exact keyword matches. A query about 'Democracy' is highly relevant to a result discussing 'elections, voting, rights, and parliament'. Assess the conceptual meaning. Only reject if the topic is completely out-of-domain."

3. Adjust the Presence Penalty:

We will reduce the artificial penalty applied to queries where the exact word is missing. This ensures strong semantic matches aren't pushed below the visibility threshold.

By implementing these changes, the system will start understanding the *intent* behind the

searches (like "Democracy" or "Education Reform") and will show relevant speeches even if those exact words were never spoken.